



FAIR 3D HERITAGE

*Documentation, Publication and Preservation
of the Digital 3D Heritage on the Web*

Publishing 3D Heritage Beyond the Model: Oral Provenance, Community Governance, and Revenue Distribution in a Pilot

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Abstract

As 3D heritage publication becomes more widespread, a challenge remains: how can digital exhibits represent not only the visual form of heritage objects, but also the social, historical, and community knowledge that gives those objects meaning? Participatory documentation has explored community contribution to metadata and interpretation (Srinivasan, Boast, Furner, & Becvar, 2009), and collaborative and participatory-action approaches in archaeology have shown that community engagement can be a site of shared authority and social benefit (Moser et al., 2002; McGhee, 2012). These literatures rarely integrate provenance enrichment or economic benefit for source communities. This paper presents a pilot from Zambia in which a web-based 3D exhibit functions simultaneously as an access environment, a provenance-enrichment platform, and a community-benefit mechanism (Gondwe et al., 2025). It asks two questions: how can 3D exhibits support richer provenance documentation through community participation, and what governance structures can connect digital heritage circulation to transparent community benefit?

The pilot centred on six digitised artefacts associated with Zambia's Southern Province, in an immersive online environment. The 3D format mattered not

only as display but as a means of presenting object detail that prompted identification and testimony during fieldwork. Fieldwork spanned two districts in the Gwembe Valley, across four handicraft cooperatives and 59 participants, with a national museum acting as institutional intermediary. Community members identified and contextualised artefacts through oral testimony, contributing provenance absent from existing catalogue records, consistent with calls for community-knowledge integration in digital heritage systems (Cameron & Kenderdine, 2007). The end-to-end workflow is summarised in Figure 1.

Ticket revenue was routed through a smart-contract escrow deployed on a public blockchain (the RVS-m contract suite). In practice, each artefact is registered on-chain with provenance metadata, tickets are issued as tokens, and revenue accrues to an escrow contract that enforces an 80/20 split favouring linked communities. Funds release only when an institutional intermediary confirms agreed deliverables on-chain. Cooperatives submitted ranked proposals, with allocation following a tranche-based structure tied to milestones. This makes fund flows auditable and tamper-resistant without requiring communities to trust a single administrator (Wilkinson et al., 2016).

The pilot produced four results. Participants recognised two of the six artefacts and contributed oral histories that expanded available provenance, including use and construction knowledge not present in existing records. Existing cooperative governance roles proved compatible with proposal-based allocation, without new local institutions. The full workflow, from exhibit publication through field engagement, escrow funding and tranche release, was confirmed operational. The case shows that FAIR-aligned 3D publication can extend beyond findability and access when structured object records, persistent identifiers, and documented community contributions form a reusable workflow (Amico & Felicetti, 2024).

The study argues that 3D exhibits should be treated not as endpoints for model display, but as publication environments in which provenance, interpretation, governance, and public engagement are brought into relation (Sarr & Savoy, 2018). The escrow, cooperative-proposal, and oral-provenance components are modular and not dependent on the Zambian context, and so speak directly to best documentation practices, web-based publication infrastructures, and public engagement with 3D.

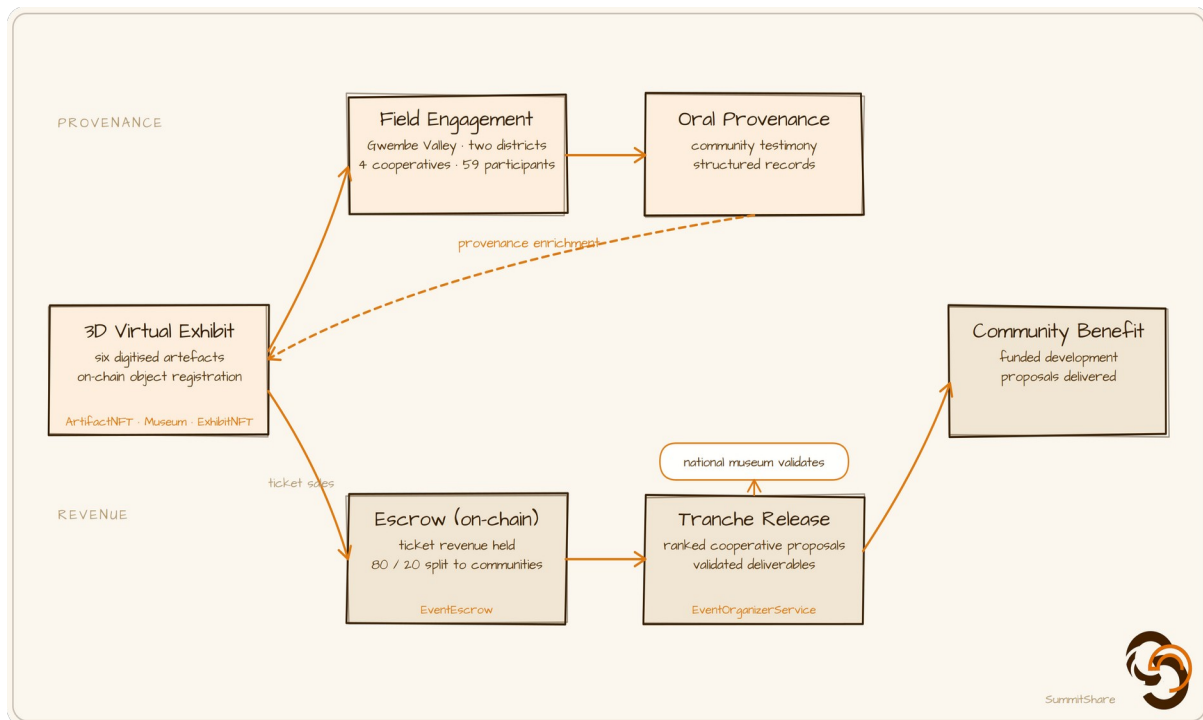


Fig. 1 The SummitShare exhibit publication workflow (RVS-m contract suite). A shared 3D exhibit feeds two coupled flows: on-chain provenance records, created through the tokenisation of artefacts and their metadata; and an escrow distribution system tied to ticket revenue. Both are facilitated by the contract mesh.

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